Basic App Check Setup Worksheet

What this is: An editable worksheet for every implemented IOS and IOS-XE L2 and NDM check. It tells you what the app looks for, whether you must enter a site value, where that value goes, and what evidence to review.

What this is not: A completed CKL or an approval decision. Confirm results against your current benchmark, device evidence, and site policy. The separate Complete Check Reference gives the exact expressions and all decision details.

How to use this worksheet

1. Keep this Word file as your working copy. Search a V number with Ctrl+F, then replace each [enter …] prompt with the approved value or reviewer note.
2. On the Audit tab, choose IOS or IOS XE, L2 or NDM (or both), and the site profile. Live SSH requires the target addresses and SSH credentials. Those connection inputs apply to the whole audit and are not repeated under every check.
3. For each check marked “Site values required”, its card below names the exact YAML field and default file. IOS uses Application\data\profiles\base_ios_switch_access.yaml; IOS XE uses Application\data\profiles\base_iosxe_access.yaml. If Audit selects a different profile, edit that selected file instead. Close and reopen the app after adding or renaming profiles. Preserve YAML indentation and list syntax.
4. “No site value” means nothing is entered in profile YAML for that check. The app compares device output with a built-in rule in Application\data\checks. Do not edit check YAML for ordinary site setup. For example, the ACL deny logging check examines every deny statement in interface-bound IPv4 ACLs and requires log-input on the deny statement; log-input is a device configuration keyword, not a YAML value. “Manual review” requires separate evidence.
5. After a run, use Results > Open protected report to view the encrypted .txt.sapenc report inside Basic. To review a CKL in STIG Viewer, deliberately export a readable .ckl copy; it is unencrypted and must be handled under site policy. Basic writes Error and Skipped as Not_Reviewed in CKL. Retain redacted evidence and record any manual decision separately.

Important file path: Application\data\profiles\<selected profile>.yaml is beside the EXE in the extracted package. IOS normally uses base_ios_switch_access.yaml; IOS XE normally uses base_iosxe_access.yaml. Edit the profile you actually selected on Audit.

Profile caution: The supplied values are examples from the original installation, including server names and IPs. Replace them with values approved for your site before a live assessment.

Three worked examples

One string - IOS XE: Open Application\data\profiles\base_iosxe_access.yaml. Find management_access: and its acl_name: line. Replace the text after acl_name: with the exact approved ACL name used on the switch. This is an example of the YAML position, not a value to use in a live audit:

```
management_access:
  acl_name: EXAMPLE-MGMT-ACL
```

Several strings - IOS: Open Application\data\profiles\base_ios_switch_access.yaml. Under variables:, find logging_servers:. Put each approved logging host on its own indented hyphen line. Replace these example names with the exact IP addresses or names after logging host in the site's switch configuration:

```
variables:
  logging_servers:
    - log-a.example.invalid
    - log-b.example.invalid
```

No YAML input - ACL deny logging: For IOS XE V-220529 or IOS V-220581, leave the profile alone. The app reads show running-config, finds interface-bound IPv4 ACLs, and looks for log-input on each deny statement. The keyword does not have to be the last token. If the device has no matching deny statements, inspect the ACL scope yourself before accepting the result.

Site values to gather before testing

The cards below identify the exact YAML field to edit and the IOS or IOS XE base file. Enter the approved value in the file selected on Audit; a worksheet answer alone does not change the app. For checks with no input, the card says what the app looks for and which check YAML defines that rule.

Profile field	What to obtain from the site owner	Your approved value
arp_inspection.vlans	User VLAN numbers expected to have Dynamic ARP	[enter value]

	Inspection configured.	
dhcp_snooping.vlans	User VLAN numbers expected to have DHCP snooping configured.	[enter value]
endpoint_authentication.radius_group	AAA RADIUS server group name as configured on the switch.	[enter value]
endpoint_authentication.radius_server_addresses	Mapping of each listed RADIUS configuration name to its expected IPv4 address.	[enter value]
endpoint_authentication.radius_servers	List of required radius server names; use the configuration names, not necessarily DNS hostnames.	[enter value]
management_access.acl_name	Name of the ACL applied inbound to VTY management lines.	[enter value]
management_access.networks	Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.	[enter value]
native_vlan	Approved native VLAN number on trunks.	[enter value]
root_guard.upstream_switches	Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.	[enter value]
trunk_policy.additional_pruned_vlans	Additional VLAN numbers forbidden on trunks, where referenced by a check.	[enter value]
unused_vlan	VLAN reserved for unused access ports. Ask the network owner for the approved number.	[enter value]
variables.local_password_policy	Name of the aaa common-criteria policy referenced by local user configuration.	[enter value]
variables.logging_servers	Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.	[enter value]
variables.max_concurrent_management_sessions	Organization-approved maximum concurrent management sessions, a positive integer.	[enter value]
variables.ntp_servers	Exact IP addresses or hostnames following ntp server. List at least two approved sources.	[enter value]
variables.supported_ios_versions	Exact approved IOS release strings from show version. The shipped IOS value is 15.2(7)E14; the organization must maintain the list.	[enter value]
variables.supported_iosxe_versions	Exact approved IOS-XE release strings from show version. The shipped value is 17.12.08; the organization must maintain the list.	[enter value]
variables.tacacs_group	TACACS+ server group name used by the default login method list.	[enter value]

IOS Layer 2 checks

How to read the cards: “What the app checks” summarizes its implemented test; the Complete Check Reference contains exact logic. Replace the bracketed worksheet prompts in Word. Apply named site values in the selected YAML profile.

V-220623 The Cisco switch must uniquely identify and authenticate all network-connected endpoint devices before establishing any connection.

Setup type: Site values required

What the app checks: Require 802.1X and MAB on every physical access port and require all profile-defined RADIUS servers in the profile-defined AAA server group

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at endpoint_authentication.radius_group. If you selected another profile on Audit, edit that profile instead. AAA RADIUS server group name as configured on the switch.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at endpoint_authentication.radius_servers. If you selected another profile on Audit, edit that profile instead. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220624 The Cisco switch must authenticate all VLAN Trunk Protocol (VTP) messages with a hash function using the most secured cryptographic algorithm available.

Setup type: No site value

What the app checks: Looking to see if VTP status is off in "show vtp status"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show vtp status, show running-config and applies the built-in rule: Looking to see if VTP status is off in "show vtp status". The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show vtp status, show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220625 The Cisco switch must manage excess bandwidth to limit the effects of packet-flooding types of denial-of-service (DoS) attacks.

Setup type: No site value

What the app checks: Require the global mls qos command in show running-config

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Require the global mls qos command in show running-config. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220629 The Cisco switch must have Root Guard enabled on all switch ports connecting to access layer switches.

Setup type: Site values required

What the app checks: Use CDP to identify connected switches. Exempt core/distribution neighbors containing 9500 or 9600, and require spanning-tree guard root on access-switch-facing ports.

How it checks: It uses CDP neighbors to identify switch-facing links, exempts the approved upstream Device IDs, and checks Root Guard on other applicable local links.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at root_guard.upstream_switches. If you selected another profile on Audit, edit that profile instead. Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show cdp neighbors detail

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220630 The Cisco switch must have Bridge Protocol Data Unit (BPDU) Guard enabled on all user-facing or untrusted access switch ports.

Setup type: No site value

What the app checks: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220631 The Cisco switch must have Spanning Tree Protocol (STP) Loop Guard enabled.

Setup type: No site value

What the app checks: Looking for list of strings in "show run"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run". The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220632 The Cisco switch must have Unknown Unicast Flood Blocking (UUFB) enabled.

Setup type: No site value

What the app checks: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports.

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220633 The Cisco switch must have DHCP snooping for all user VLANs to validate DHCP messages from untrusted sources.

Setup type: Site values required

What the app checks: Ensure global DHCP snooping and all profile-defined user VLANs are configured

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at dhcp_snooping.vlans. If you selected another profile on Audit, edit that profile instead. User VLAN numbers expected to have DHCP snooping configured.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220634 The Cisco switch must have IP Source Guard enabled on all user-facing or untrusted access switch ports.

Setup type: No site value

What the app checks: Look through all access ports to ensure that "ip verify source" is configured on all access ports.

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Look through all access ports to ensure that "ip verify source" is configured on all access ports. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220635 The Cisco switch must have Dynamic Address Resolution Protocol (ARP) Inspection (DAI) enabled on all user VLANs.

Setup type: Site values required

What the app checks: Ensure Dynamic ARP Inspection is configured for all profile-defined user VLANs

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at arp_inspection.vlans. If you selected another profile on Audit, edit that profile instead. User VLAN numbers expected to have Dynamic ARP Inspection configured.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220636 The Cisco switch must have Storm Control configured on all host-facing switchports.

Setup type: No site value

What the app checks: Loop through all access ports to ensure multiple strings are applied

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Loop through all access ports to ensure multiple strings are applied. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220637 The Cisco switch must have IGMP or MLD Snooping configured on all VLANs.

Setup type: No site value

What the app checks: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220638 The Cisco switch must implement Rapid Spanning Tree Protocol (STP) where VLANs span multiple switches with redundant links.

Setup type: No site value

What the app checks: Looking for list of strings in "show run"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run". The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220639 The Cisco switch must enable Unidirectional Link Detection (UDLD) to protect against one-way connections.

Setup type: No site value

What the app checks: Looking for list of strings in "show run | inc udld"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run | inc udld". The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220640 The Cisco switch must have all trunk links enabled statically.

Setup type: No site value

What the app checks: Ensure all switchports report Negotiation of Trunking as Off

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show interfaces switchport | include Negotiation of Trunking and applies the built-in rule: Ensure all switchports report Negotiation of Trunking as Off. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show interfaces switchport | include Negotiation of Trunking

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220641 The Cisco switch must have all disabled switch ports assigned to an unused VLAN.

Setup type: Site values required

What the app checks: Exempt 802.1X access ports; require all other disabled or disconnected access ports to be shut down and assigned to the unused VLAN

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at unused_vlan. If you selected another profile on Audit, edit that profile instead. VLAN reserved for unused access ports. Ask the network owner for the approved number.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220642 The Cisco switch must not have the default VLAN assigned to any host-facing switch ports.

Setup type: No site value

What the app checks: Loop through access ports to make sure none are assigned to VLAN 1

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Loop through access ports to make sure none are assigned to VLAN 1. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220643 The Cisco switch must have the default VLAN pruned from all trunk ports that do not require it.

Setup type: Site values required

What the app checks: Loop through trunk ports to ensure all have VLAN 1 pruned

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at trunk_policy.additional_pruned_vlans. If you selected another profile on Audit, edit that profile instead. Additional VLAN numbers forbidden on trunks, where referenced by a check.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220644 The Cisco switch must not use the default VLAN for management traffic.

Setup type: No site value

What the app checks: Ensure VLAN 1 is not the management VLAN and has no IP address assigned

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure VLAN 1 is not the management VLAN and has no IP address assigned. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220645 The Cisco switch must have all user-facing or untrusted ports configured as access switch ports.

Setup type: No site value

What the app checks: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220646 The Cisco switch must have the native VLAN assigned to an ID other than the default VLAN for all 802.1q trunk links.

Setup type: No site value

What the app checks: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1. The exact expressions are in Application\data\checks\ios_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220647 The Cisco switch must not have any switchports assigned to the native VLAN.

Setup type: Site values required

What the app checks: Ensure no physical switchport is assigned as an access port in the profile-defined native VLAN

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at native_vlan. If you selected another profile on Audit, edit that profile instead. Approved native VLAN number on trunks.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

IOS Network Device Management checks

How to read the cards: “What the app checks” summarizes its implemented test; the Complete Check Reference contains exact logic. Replace the bracketed worksheet prompts in Word. Apply named site values in the selected YAML profile.

V-220570 The Cisco switch must be configured to limit the number of concurrent management sessions to an organization-defined number.

Setup type: Site values required

What the app checks: Validate session-limit under every VTY section, or count the available VTY lines when session-limit is unsupported

How it checks: It checks configured session-limit values where present, otherwise counts configured VTY lines against the site maximum.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.max_concurrent_management_sessions. If you selected another profile on Audit, edit that profile instead. Organization-approved maximum concurrent management sessions, a positive integer.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220575 The Cisco switch must be configured to enforce approved authorizations for controlling the flow of management information within the device based on contr…

Setup type: Site values required

What the app checks: Require the profile-defined management ACL inbound on every VTY section and permit only profile-approved management networks

How it checks: It checks VTY inbound access-class use and compares permits in the named ACL with the approved source networks.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at management_access.acl_name. If you selected another profile on Audit, edit that profile instead. Name of the ACL applied inbound to VTY management lines.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at management_access.networks. If you selected another profile on Audit, edit that profile instead. Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220576 The Cisco switch must be configured to enforce the limit of three consecutive invalid logon attempts, after which time it must lock out the user account f…

Setup type: No site value

What the app checks: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220577 The Cisco switch must be configured to display the Standard Mandatory DoD Notice and Consent Banner before granting access to the device.

Setup type: No site value

What the app checks: Verify the login banner contains every mandatory DoD notice and consent clause

How it checks: It parses the login banner and looks for the built-in DoD notice and consent clauses.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Verify the login banner contains every mandatory DoD notice and consent clause. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220578 The Cisco device must be configured to audit all administrator activity.

Setup type: No site value

What the app checks: Ensure administrator configuration changes generate informational logging messages

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure administrator configuration changes generate informational logging messages. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220580 The Cisco switch must produce audit records containing information to establish when (date and time) the events occurred.

Setup type: No site value

What the app checks: Require service timestamps log datetime msec localtime, allowing additional options afterward

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Require service timestamps log datetime msec localtime, allowing additional options afterward. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220581 The Cisco switch must produce audit records containing information to establish where the events occurred.

Setup type: No site value

What the app checks: Require log-input on every deny statement in each interface-bound IPv4 ACL

How it checks: It parses interface-bound IPv4 standard and extended ACLs and requires log-input on each selected deny statement. IPv6 ACLs are outside this implemented scope.

YAML input: None. Do not add log-input to a profile YAML file. The app reads show running-config, finds every interface-bound IPv4 ACL, and looks for log-input anywhere on each deny statement; it does not require the keyword to be the final token. If no matching deny statements are found, review the ACL scope manually. The rule is in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show ip access-lists

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220583 The Cisco switch must be configured to protect audit information from unauthorized modification.

Setup type: No site value

What the app checks: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

How it checks: It checks whether persistent logging is configured; when present, it evaluates the configured file privilege value.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220584 The Cisco switch must be configured to protect audit information from unauthorized deletion.

Setup type: No site value

What the app checks: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

How it checks: It checks whether persistent logging is configured; when present, it evaluates the configured file privilege value.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220585 The Cisco switch must be configured to limit privileges to change the software resident within software libraries.

Setup type: No site value

What the app checks: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value.

How it checks: It checks whether persistent logging is configured; when present, it evaluates the configured file privilege value.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: If logging persistent is absent, mark Not Applicable. If enabled, require file privilege 15 and no other file privilege value. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220586 The Cisco switch must be configured to prohibit the use of all unnecessary and non-secure functions and services.

Setup type: No site value

What the app checks: Ensure prohibited and unnecessary services are not enabled in the running configuration

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure prohibited and unnecessary services are not enabled in the running configuration. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220587 The Cisco switch must be configured with only one local account to be used as the account of last resort in the event the authentication server is unavail…

Setup type: Site values required

What the app checks: Require exactly one local username using the profile-defined common-criteria password policy and require the profile-defined TACACS group before local in the default login authentication order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.tacacs_group. If you selected another profile on Audit, edit that profile instead. TACACS+ server group name used by the default login method list.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220589 The Cisco switch must be configured to enforce a minimum 15-character password length.

Setup type: Site values required

What the app checks: Require min-length 15 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220590 The Cisco switch must be configured to enforce password complexity by requiring that at least one uppercase character be used.

Setup type: Site values required

What the app checks: Require upper-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220591 The Cisco switch must be configured to enforce password complexity by requiring that at least one lowercase character be used.

Setup type: Site values required

What the app checks: Require lower-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220592 The Cisco switch must be configured to enforce password complexity by requiring that at least one numeric character be used.

Setup type: Site values required

What the app checks: Require numeric-count 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220593 The Cisco switch must be configured to enforce password complexity by requiring that at least one special character be used.

Setup type: Site values required

What the app checks: Require special-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220594 The Cisco switch must be configured to require that when a password is changed, the characters are changed in at least eight of the positions within the p…

Setup type: Site values required

What the app checks: Require char-changes 8 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220595 The Cisco switch must only store cryptographic representations of passwords.

Setup type: No site value

What the app checks: Ensure service password-encryption is enabled

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure service password-encryption is enabled. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220596 The Cisco switch must be configured to terminate all network connections associated with device management after five minutes of inactivity.

Setup type: No site value

What the app checks: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220599 The Cisco switch must be configured to allocate audit record storage capacity in accordance with organization-defined audit record storage requirements.

Setup type: No site value

What the app checks: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220600 The Cisco switch must be configured to generate an alert for all audit failure events.

Setup type: No site value

What the app checks: Ensure remote logging uses a valid named trap severity level

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure remote logging uses a valid named trap severity level. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220601 The Cisco switch must be configured to synchronize its clock with the primary and secondary time sources using redundant authoritative time sources.

Setup type: Site values required

What the app checks: Require at least two NTP server commands and every NTP server defined in the selected profile

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.ntp_servers. If you selected another profile on Audit, edit that profile instead. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220604 The Cisco switch must be configured to authenticate SNMP messages using a FIPS-validated Keyed-Hash Message Authentication Code (HMAC).

Setup type: No site value

What the app checks: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show snmp user and applies the built-in rule: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show snmp user

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220605 The Cisco switch must be configured to encrypt SNMP messages using a FIPS 140-2 approved algorithm.

Setup type: No site value

What the app checks: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show snmp user and applies the built-in rule: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show snmp user

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220606 The Cisco switch must be configured to authenticate Network Time Protocol (NTP) sources using authentication with FIPS-compliant algorithms.

Setup type: No site value

What the app checks: Flag MD5 NTP authentication keys as Open; review any other configured algorithm for FIPS compliance

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Flag MD5 NTP authentication keys as Open; review any other configured algorithm for FIPS compliance. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220607 The Cisco switch must be configured to use FIPS-validated Keyed-Hash Message Authentication Code (HMAC) to protect the integrity of remote maintenance ses…

Setup type: No site value

What the app checks: Require the approved SSH HMAC algorithms in the configured order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config | include ssh and applies the built-in rule: Require the approved SSH HMAC algorithms in the configured order. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config | include ssh

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220608 The Cisco switch must be configured to implement cryptographic mechanisms to protect the confidentiality of remote maintenance sessions.

Setup type: No site value

What the app checks: Require the approved SSH AES-CTR encryption algorithms in the configured order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config | include ssh and applies the built-in rule: Require the approved SSH AES-CTR encryption algorithms in the configured order. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config | include ssh

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220609 The Cisco switch must be configured to protect against known types of denial-of-service (DoS) attacks by employing organization-defined security safeguard…

Setup type: Manual review

What the app checks: C-22324r1007810_chk Review the Cisco switch configuration to verify that it protects against known types of DoS attacks by employing organization-defined security safeguards. Note: Cisco IOS Version 15 and newer could use the command "mls qos copp protocol" to meet the intent of this requ…

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220612 The Cisco switch must be configured to generate audit records when successful/unsuccessful logon attempts occur.

Setup type: No site value

What the app checks: Ensure successful and failed login attempts are logged

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_ios_switch_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure successful and failed login attempts are logged. The exact expressions are in Application\data\checks\ios_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220617 The Cisco switch must be configured to use at least two authentication servers to authenticate users prior to granting administrative access.

Setup type: Site values required

What the app checks: Require at least two profile-defined RADIUS server blocks with the correct address, authentication port, accounting port, key, and AAA group membership

How it checks: It checks the configured RADIUS group, each required server name/address, ports 1812 and 1813, and a key command. It does not log in to the RADIUS server.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at endpoint_authentication.radius_group. If you selected another profile on Audit, edit that profile instead. AAA RADIUS server group name as configured on the switch.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at endpoint_authentication.radius_server_addresses. If you selected another profile on Audit, edit that profile instead. Mapping of each listed RADIUS configuration name to its expected IPv4 address.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at endpoint_authentication.radius_servers. If you selected another profile on Audit, edit that profile instead. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220618 The Cisco switch must be configured to support organizational requirements to conduct backups of the configuration when changes occur.

Setup type: Manual review

What the app checks: Temporarily mark NotAFinding because configuration backups are performed through Cisco Catalyst Center

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220619 The Cisco switch must be configured to obtain its public key certificates from an appropriate certificate policy through an approved service provider.

Setup type: Manual review

What the app checks: Temporarily Not Applicable: only non-removable Cisco default certificates are present; revisit certificate applicability later

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220620 The Cisco switch must be configured to send log data to at least two central log servers for the purpose of forwarding alerts to the administrators and th…

Setup type: Site values required

What the app checks: Require at least two logging host commands and every logging server defined in the selected profile

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.logging_servers. If you selected another profile on Audit, edit that profile instead. Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220621 The Cisco switch must be running an IOS release that is currently supported by Cisco Systems.

Setup type: Site values required

What the app checks: Match the installed IOS release against the profile-defined approved version

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_ios_switch_access.yaml at variables.supported_ios_versions. If you selected another profile on Audit, edit that profile instead. Exact approved IOS release strings from show version. The shipped IOS value is 15.2(7)E14; the organization must maintain the list.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show version

Your notes: [enter local example, expected result, issue, or assessor follow-up]

IOS XE Layer 2 checks

How to read the cards: “What the app checks” summarizes its implemented test; the Complete Check Reference contains exact logic. Replace the bracketed worksheet prompts in Word. Apply named site values in the selected YAML profile.

V-220649 The Cisco switch must uniquely identify and authenticate all network-connected endpoint devices before establishing any connection.

Setup type: Site values required

What the app checks: Require 802.1X and MAB on every physical access port and require all profile-defined RADIUS servers in the profile-defined AAA server group

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at endpoint_authentication.radius_group. If you selected another profile on Audit, edit that profile instead. AAA RADIUS server group name as configured on the switch.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at endpoint_authentication.radius_servers. If you selected another profile on Audit, edit that profile instead. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220650 The Cisco switch must authenticate all VLAN Trunk Protocol (VTP) messages with a hash function using the most secured cryptographic algorithm available.

Setup type: No site value

What the app checks: Looking to see if VTP status is off in "show vtp status"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show vtp status, show running-config and applies the built-in rule: Looking to see if VTP status is off in "show vtp status". The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show vtp status, show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220651 The Cisco switch must manage excess bandwidth to limit the effects of packet flooding types of denial-of-service (DoS) attacks.

Setup type: Manual review

What the app checks: QoS is not currently configured; retain as Open until the site QoS policy is implemented and this check is automated

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220655 The Cisco switch must have Root Guard enabled on all switch ports connecting to access layer switches.

Setup type: Site values required

What the app checks: Use CDP to identify directly connected switches. Exempt configured core/distribution neighbors and require "spanning-tree guard root" on all other switch-facing interfaces.

How it checks: It uses CDP neighbors to identify switch-facing links, exempts the approved upstream Device IDs, and checks Root Guard on other applicable local links.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at root_guard.upstream_switches. If you selected another profile on Audit, edit that profile instead. Exact upstream core/distribution Device IDs from show cdp neighbors detail. Short names and fully qualified names are matched. These links are exempt from Root Guard.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show cdp neighbors detail

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220656 The Cisco switch must have BPDU Guard enabled on all user-facing or untrusted access switch ports.

Setup type: No site value

What the app checks: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Check if "spanning-tree bpduguard enable" is set globally, and if not, make sure it is applied to all access ports. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220657 The Cisco switch must have STP Loop Guard enabled.

Setup type: No site value

What the app checks: Looking for list of strings in "show run"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run". The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220658 The Cisco switch must have Unknown Unicast Flood Blocking (UUFB) enabled.

Setup type: No site value

What the app checks: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports.

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Look through all access ports to ensure that "switchport block unicast" is configured on all access ports. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220659 The Cisco switch must have DHCP snooping for all user VLANs to validate DHCP messages from untrusted sources.

Setup type: Site values required

What the app checks: Ensure global DHCP snooping and all profile-defined user VLANs are configured

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at dhcp_snooping.vlans. If you selected another profile on Audit, edit that profile instead. User VLAN numbers expected to have DHCP snooping configured.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220660 The Cisco switch must have IP Source Guard enabled on all user-facing or untrusted access switch ports.

Setup type: No site value

What the app checks: Look through all access ports to ensure that "ip verify source" is configured on all access ports.

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Look through all access ports to ensure that "ip verify source" is configured on all access ports. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220661 The Cisco switch must have Dynamic Address Resolution Protocol (ARP) Inspection (DAI) enabled on all user VLANs.

Setup type: Site values required

What the app checks: Ensure Dynamic ARP Inspection is configured for all profile-defined user VLANs

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at arp_inspection.vlans. If you selected another profile on Audit, edit that profile instead. User VLAN numbers expected to have Dynamic ARP Inspection configured.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220662 The Cisco switch must have Storm Control configured on all host-facing switchports.

Setup type: No site value

What the app checks: Loop through all access ports to ensure multiple strings are applied

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Loop through all access ports to ensure multiple strings are applied. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220663 The Cisco switch must have IGMP or MLD Snooping configured on all VLANs.

Setup type: No site value

What the app checks: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure IGMP and IPv6 MLD snooping are not disabled globally or for individual VLANs. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220664 The Cisco switch must implement Rapid STP where VLANs span multiple switches with redundant links.

Setup type: No site value

What the app checks: Looking for list of strings in "show run"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run". The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220665 The Cisco switch must enable Unidirectional Link Detection (UDLD) to protect against one-way connections.

Setup type: No site value

What the app checks: Looking for list of strings in "show run | inc udld"

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Looking for list of strings in "show run | inc udld". The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220666 The Cisco switch must have all trunk links enabled statically.

Setup type: No site value

What the app checks: Ensure all switchports report Negotiation of Trunking as Off

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show interfaces switchport | include Negotiation of Trunking and applies the built-in rule: Ensure all switchports report Negotiation of Trunking as Off. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show interfaces switchport | include Negotiation of Trunking

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220667 The Cisco switch must have all disabled switch ports assigned to an unused VLAN.

Setup type: Site values required

What the app checks: Exempt 802.1X access ports; require all other disabled or disconnected access ports to be shut down and assigned to the unused VLAN

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at unused_vlan. If you selected another profile on Audit, edit that profile instead. VLAN reserved for unused access ports. Ask the network owner for the approved number.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220668 The Cisco switch must not have the default VLAN assigned to any host-facing switch ports.

Setup type: No site value

What the app checks: Loop through access ports to make sure none are assigned to VLAN 1

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Loop through access ports to make sure none are assigned to VLAN 1. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220669 The Cisco switch must have the default VLAN pruned from all trunk ports that do not require it.

Setup type: Site values required

What the app checks: Loop through trunk ports to ensure all have VLAN 1 pruned

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at trunk_policy.additional_pruned_vlans. If you selected another profile on Audit, edit that profile instead. Additional VLAN numbers forbidden on trunks, where referenced by a check.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220670 The Cisco switch must not use the default VLAN for management traffic.

Setup type: No site value

What the app checks: Ensure VLAN 1 is not the management VLAN and has no IP address assigned

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure VLAN 1 is not the management VLAN and has no IP address assigned. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220671 The Cisco switch must have all user-facing or untrusted ports configured as access switch ports.

Setup type: No site value

What the app checks: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Require switchport mode access on every physical copper Ethernet port; ignore SFP/fiber ports used as uplinks. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220672 The Cisco switch must have the native VLAN assigned to an ID other than the default VLAN for all 802.1q trunk links.

Setup type: No site value

What the app checks: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config, show interfaces status and applies the built-in rule: Ensure every physical trunk has an explicitly configured native VLAN other than VLAN 1. The exact expressions are in Application\data\checks\iosxe_l2.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220673 The Cisco switch must not have any switchports assigned to the native VLAN.

Setup type: Site values required

What the app checks: Ensure no physical switchport is assigned as an access port in the profile-defined native VLAN

How it checks: It examines each interface in the configured scope for required and forbidden commands. An applicable interface without a required command can fail.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at native_vlan. If you selected another profile on Audit, edit that profile instead. Approved native VLAN number on trunks.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show interfaces status

Your notes: [enter local example, expected result, issue, or assessor follow-up]

IOS XE Network Device Management checks

How to read the cards: “What the app checks” summarizes its implemented test; the Complete Check Reference contains exact logic. Replace the bracketed worksheet prompts in Word. Apply named site values in the selected YAML profile.

V-220518 The Cisco switch must be configured to limit the number of concurrent management sessions to an organization-defined number.

Setup type: Site values required

What the app checks: Validate session-limit under every VTY section, or count the available VTY lines when session-limit is unsupported

How it checks: It checks configured session-limit values where present, otherwise counts configured VTY lines against the site maximum.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.max_concurrent_management_sessions. If you selected another profile on Audit, edit that profile instead. Organization-approved maximum concurrent management sessions, a positive integer.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220519 The Cisco switch must be configured to automatically audit account creation.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220520 The Cisco switch must be configured to automatically audit account modification.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220521 The Cisco switch must be configured to automatically audit account disabling actions.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220522 The Cisco switch must be configured to automatically audit account removal actions.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220523 The Cisco switch must be configured to enforce approved authorizations for controlling the flow of management information within the device based on contr…

Setup type: Site values required

What the app checks: Require the profile-defined management ACL inbound on every VTY section and permit only profile-approved management networks

How it checks: It checks VTY inbound access-class use and compares permits in the named ACL with the approved source networks.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at management_access.acl_name. If you selected another profile on Audit, edit that profile instead. Name of the ACL applied inbound to VTY management lines.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at management_access.networks. If you selected another profile on Audit, edit that profile instead. Approved source networks as network_address and subnet_mask. Use an ordinary subnet mask, not an ACL wildcard mask.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220524 The Cisco switch must be configured to enforce the limit of three consecutive invalid logon attempts, after which time it must lock out the user account f…

Setup type: No site value

What the app checks: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure login blocking is configured for 900 seconds after three failed attempts within 120 seconds. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220525 The Cisco switch must be configured to display the Standard Mandatory DoD Notice and Consent Banner before granting access to the device.

Setup type: No site value

What the app checks: Verify the login banner contains every mandatory DoD notice and consent clause

How it checks: It parses the login banner and looks for the built-in DoD notice and consent clauses.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Verify the login banner contains every mandatory DoD notice and consent clause. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220526 The Cisco device must be configured to audit all administrator activity.

Setup type: No site value

What the app checks: Ensure administrator configuration changes generate informational logging messages

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure administrator configuration changes generate informational logging messages. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220528 The Cisco switch must produce audit records containing information to establish when (date and time) the events occurred.

Setup type: No site value

What the app checks: Ensure log messages use millisecond local date and time timestamps with timezone and year

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure log messages use millisecond local date and time timestamps with timezone and year. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Current rule: The IOS-XE V-220528 timestamp expression requires service timestamps log datetime msec localtime show-timezone year as a full line. A missing year or extra token after year fails; compare any finding with the device configuration.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220529 The Cisco switch must produce audit records containing information to establish where the events occurred.

Setup type: No site value

What the app checks: Require log-input on every deny statement in each interface-bound IPv4 ACL

How it checks: It parses interface-bound IPv4 standard and extended ACLs and requires log-input on each selected deny statement. IPv6 ACLs are outside this implemented scope.

YAML input: None. Do not add log-input to a profile YAML file. The app reads show running-config, finds every interface-bound IPv4 ACL, and looks for log-input anywhere on each deny statement; it does not require the keyword to be the final token. If no matching deny statements are found, review the ACL scope manually. The rule is in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config, show ip access-lists

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220530 The Cisco switch must be configured to generate audit records containing the full-text recording of privileged commands.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220531 The Cisco switch must be configured to protect audit information from unauthorized modification.

Setup type: Manual review

What the app checks: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220532 The Cisco switch must be configured to protect audit information from unauthorized deletion.

Setup type: Manual review

What the app checks: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220533 The Cisco switch must be configured to limit privileges to change the software resident within software libraries.

Setup type: Manual review

What the app checks: Logging persistent is not configured; temporarily Not Applicable pending future implementation and validation

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220534 The Cisco switch must be configured to prohibit the use of all unnecessary and nonsecure functions and services.

Setup type: No site value

What the app checks: Ensure prohibited and unnecessary services are not enabled in the running configuration

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure prohibited and unnecessary services are not enabled in the running configuration. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220535 The Cisco switch must be configured with only one local account to be used as the account of last resort in the event the authentication server is unavail…

Setup type: Site values required

What the app checks: Require exactly one local username using the profile-defined common-criteria password policy and require the profile-defined TACACS group before local in the default login authentication order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.tacacs_group. If you selected another profile on Audit, edit that profile instead. TACACS+ server group name used by the default login method list.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220537 The Cisco switch must be configured to enforce a minimum 15-character password length.

Setup type: Site values required

What the app checks: Require min-length 15 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220538 The Cisco switch must be configured to enforce password complexity by requiring that at least one uppercase character be used.

Setup type: Site values required

What the app checks: Require upper-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220539 The Cisco switch must be configured to enforce password complexity by requiring that at least one lowercase character be used.

Setup type: Site values required

What the app checks: Require lower-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220540 The Cisco switch must be configured to enforce password complexity by requiring that at least one numeric character be used.

Setup type: Site values required

What the app checks: Require numeric-count 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220541 The Cisco switch must be configured to enforce password complexity by requiring that at least one special character be used.

Setup type: Site values required

What the app checks: Require special-case 1 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220542 The Cisco switch must be configured to require that when a password is changed, the characters are changed in at least eight of the positions within the p…

Setup type: Site values required

What the app checks: Require char-changes 8 under the profile-defined AAA common-criteria password policy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.local_password_policy. If you selected another profile on Audit, edit that profile instead. Name of the aaa common-criteria policy referenced by local user configuration.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220543 The Cisco switch must only store cryptographic representations of passwords.

Setup type: No site value

What the app checks: Ensure service password-encryption is enabled

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure service password-encryption is enabled. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220544 The Cisco switch must be configured to terminate all network connections associated with device management after five minutes of inactivity.

Setup type: No site value

What the app checks: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Require session-timeout 5 or exec-timeout 5 0 under every VTY line section. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220545 The Cisco switch must be configured to automatically audit account enabling actions.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220547 The Cisco switch must be configured to allocate audit record storage capacity in accordance with organization-defined audit record storage requirements.

Setup type: No site value

What the app checks: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure logging buffered specifies a numeric buffer size and a valid logging severity level. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220548 The Cisco switch must be configured to generate an alert for all audit failure events.

Setup type: No site value

What the app checks: Ensure remote logging uses a valid named trap severity level

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure remote logging uses a valid named trap severity level. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220549 The Cisco switch must be configured to synchronize its clock with the primary and secondary time sources using redundant authoritative time sources.

Setup type: Site values required

What the app checks: Require at least two NTP server commands and every NTP server defined in the selected profile

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.ntp_servers. If you selected another profile on Audit, edit that profile instead. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220552 The Cisco switch must be configured to authenticate SNMP messages using a FIPS-validated Keyed-Hash Message Authentication Code (HMAC).

Setup type: No site value

What the app checks: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show snmp user and applies the built-in rule: Require at least one SNMPv3 user using SHA or a supported SHA-2 authentication protocol. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show snmp user

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220553 The Cisco switch must be configured to encrypt SNMP messages using a FIPS 140-2 approved algorithm.

Setup type: No site value

What the app checks: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show snmp user and applies the built-in rule: Require at least one SNMPv3 user using AES-128, AES-192, or AES-256 privacy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show snmp user

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220554 The Cisco switch must be configured to authenticate Network Time Protocol (NTP) sources using authentication that is cryptographically based.

Setup type: Site values required

What the app checks: Require every profile-defined NTP server to use a trusted HMAC-SHA2-256 authentication key

How it checks: It checks profile-listed NTP server associations, trusted keys, and algorithms. The current IOS-XE allowlist includes hmac-sha2-256 and md5; the descriptive wording is narrower. Review this site-policy mismatch.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.ntp_servers. If you selected another profile on Audit, edit that profile instead. Exact IP addresses or hostnames following ntp server. List at least two approved sources.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220555 The Cisco switch must be configured to use FIPS-validated Keyed-Hash Message Authentication Code (HMAC) to protect the integrity of remote maintenance ses…

Setup type: No site value

What the app checks: Require the approved SSH HMAC algorithms in the configured order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config | include ssh and applies the built-in rule: Require the approved SSH HMAC algorithms in the configured order. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config | include ssh

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220556 The Cisco switch must be configured to implement cryptographic mechanisms to protect the confidentiality of remote maintenance sessions.

Setup type: No site value

What the app checks: Require the approved SSH AES-CTR encryption algorithms in the configured order

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config | include ssh and applies the built-in rule: Require the approved SSH AES-CTR encryption algorithms in the configured order. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config | include ssh

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220559 The Cisco switch must be configured to generate log records when administrator privileges are deleted.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220560 The Cisco switch must be configured to generate audit records when successful/unsuccessful logon attempts occur.

Setup type: No site value

What the app checks: Ensure successful and failed login attempts are logged

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure successful and failed login attempts are logged. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220561 The Cisco switch must be configured to generate log records for privileged activities.

Setup type: No site value

What the app checks: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: None. Do not add a string to Application\data\profiles\base_iosxe_access.yaml for this check. The app reads show running-config and applies the built-in rule: Ensure archive log configuration auditing is enabled in the correct configuration hierarchy. The exact expressions are in Application\data\checks\iosxe_ndm.yaml for reference; do not edit it for site setup.

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220565 The Cisco switch must be configured to use at least two authentication servers for the purpose of authenticating users prior to granting administrative ac…

Setup type: Site values required

What the app checks: Require at least two profile-defined RADIUS server blocks with the correct address, authentication port, accounting port, key, and AAA group membership

How it checks: It checks the configured RADIUS group, each required server name/address, ports 1812 and 1813, and a key command. It does not log in to the RADIUS server.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at endpoint_authentication.radius_group. If you selected another profile on Audit, edit that profile instead. AAA RADIUS server group name as configured on the switch.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at endpoint_authentication.radius_server_addresses. If you selected another profile on Audit, edit that profile instead. Mapping of each listed RADIUS configuration name to its expected IPv4 address.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at endpoint_authentication.radius_servers. If you selected another profile on Audit, edit that profile instead. List of required radius server names; use the configuration names, not necessarily DNS hostnames.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220566 The Cisco switch must be configured to support organizational requirements to conduct backups of the configuration when changes occur.

Setup type: Manual review

What the app checks: Temporarily mark NotAFinding because configuration backups are performed through Cisco Catalyst Center

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220567 The Cisco switch must be configured to obtain its public key certificates from an appropriate certificate policy through an approved service provider.

Setup type: Manual review

What the app checks: Not Applicable because these access switches do not use organization-managed public key certificates

How it checks: The app does not verify this item from device commands; it assigns the preset status shown below. Gather separate evidence and make the assessor decision.

YAML input: None. This is a manual preset, not a device-verified result. Do not add a profile string for it. Collect separate evidence and record the reviewer decision in the checklist.

Evidence or reviewer decision to add: [enter evidence source, date, assessor conclusion, and CKL comment]

Device evidence read: No device command; manual preset.

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220568 The Cisco switch must be configured to send log data to at least two central log servers for the purpose of forwarding alerts to the administrators and th…

Setup type: Site values required

What the app checks: Require at least two logging host commands and every logging server defined in the selected profile

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.logging_servers. If you selected another profile on Audit, edit that profile instead. Exact IP addresses or hostnames following logging host in the configuration. List at least two approved destinations.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show running-config

Your notes: [enter local example, expected result, issue, or assessor follow-up]

V-220569 The Cisco switch must be running an IOS release that is currently supported by Cisco Systems.

Setup type: Site values required

What the app checks: Match the installed IOS-XE release against the profile-defined list of approved Cisco-supported versions

How it checks: It searches the named command output for the required, alternative, and forbidden conditions defined in the check library.

YAML input: Enter the approved value in Application\data\profiles\base_iosxe_access.yaml at variables.supported_iosxe_versions. If you selected another profile on Audit, edit that profile instead. Exact approved IOS-XE release strings from show version. The shipped value is 17.12.08; the organization must maintain the list.

Your approved value: [enter value here and make the same edit in the selected YAML profile]

Device evidence read: show version

Your notes: [enter local example, expected result, issue, or assessor follow-up]

After you complete the worksheet

6. Save your edited Word copy with a site-specific filename and retain the final profile YAML alongside it. Keep credentials out of both files.
7. Run sample mode to learn the workflow, then use authorized live devices. Use separate runs for IOS versus IOS XE and for different policy profiles.
8. Compare every Open, Error, Not_Reviewed, manual preset, and special-case pass against collected evidence. Update the CKL in your approved review process.
9. If the check library or profile changes, regenerate or revise this worksheet. The exact test logic lives in Application\data\checks; this worksheet describes the packaged release.

Coverage: 121 check cards: 71 automated checks with no check-specific site value, 41 requiring site values, and 9 manual presets. IOS L2 22; IOS NDM 35; IOS XE L2 22; IOS XE NDM 42.